

1 A Nice Kernel. Let $\mathcal{X} = \mathbb{R}^d$. Define:

$$k(x_k, x_l) = \sum_{i, \lambda_i > 0} \lambda_i \phi_i(x_k) \phi_i(x_l)$$

where λ_i are the eigenvalues and $\phi_i(x)$ are the eigenfunctions of K.

1. Verify that K is pos. definite.

Let $\alpha \in \mathbb{R}^n$, then

$$\begin{aligned} \sum_{kl} \alpha_k \alpha_l K(x_k, x_l) &= \sum_{kl} \sum_i \alpha_k \alpha_l \lambda_i \phi_i(x_k) \phi_i(x_l) = \\ \sum_i \left(\sum_k \sqrt{\lambda_i} \alpha_k \phi_i(x_k) \sum_l \sqrt{\lambda_i} \alpha_l \phi_i(x_l) \right) &= \\ \sum_i (\Phi_i(x))^2 \geq 0, \text{ where } \Phi_i(x) &\equiv \sum_m \sqrt{\lambda_i} \alpha_m \phi_i(x_m) \end{aligned}$$

*fixed missing α_i s from first version.

2. Show that the Hilbert space H is RKHS.

K is in H since

$$\begin{aligned} f_x(\cdot) = K(x, \cdot) &= \sum_k \gamma_k K(x_k, \cdot) = \sum_k \gamma_k \sum_i \lambda_i \phi_i(x_k) \phi_i(\cdot) = \\ \sum_i \lambda_i \left(\sum_k \gamma_k \phi_i(x_k) \right) \phi_i(\cdot) \end{aligned}$$

Define $\beta_i = \lambda_i \left(\sum_k \gamma_k \phi_i(x_k) \right)$, then

$$f_x(\cdot) = K(x, \cdot) = \sum_i \beta_i \phi_i(\cdot) \implies K(x_i, x_j) = \sum_i \beta_i \phi_i(x_j) = f_x(x_j)$$

Thus K has the form of an $f \in H$.

3. K has the reproducing property since

$$\begin{aligned} \langle f_\alpha, K(x, \cdot) \rangle &= \langle \sum_i \alpha_i \phi_i(\cdot), \sum_i \lambda_i \phi_i(x) \phi_i(\cdot) \rangle = \\ \sum_i \alpha_i \phi_i(x) \langle \phi_i(\cdot), \phi_i(\cdot) \rangle &= \sum_i \alpha_i \phi_i(x) = f_\alpha(x) \end{aligned}$$

This follows since eigenfunctions are orthonormal. Thus H is RKHS.

4. Show that the inner product in H can be written as $\sum_i \alpha_i \beta_i$

Take $\alpha'_i = \alpha_i / \sqrt{\lambda_i}$. And redefine H with $\alpha \rightarrow \alpha'$ and inner product:

$$\sum_i \frac{\alpha_i \beta_i}{\lambda_i} = \sum_i \frac{\sqrt{\lambda_i} \alpha'_i \sqrt{\lambda_i} \beta'_i}{\lambda_i} = \sum_i \alpha'_i \beta'_i$$

*Added forgotten primes in first versions